Ambumetrix

Optimal hospital selection for EMS transport of COVID-19 patients

Team 176

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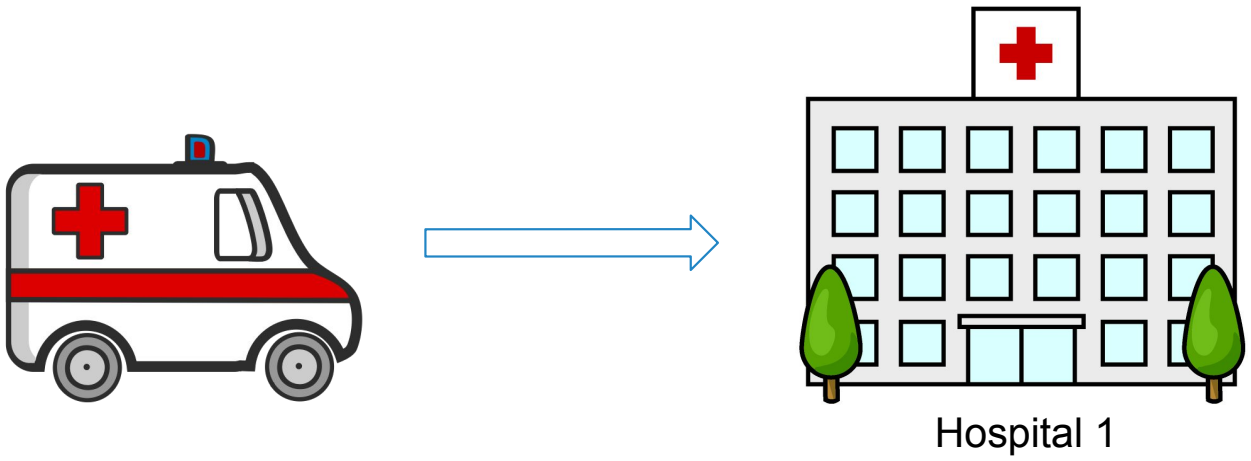
EMS routing decisions impair COVID-19 care delivery



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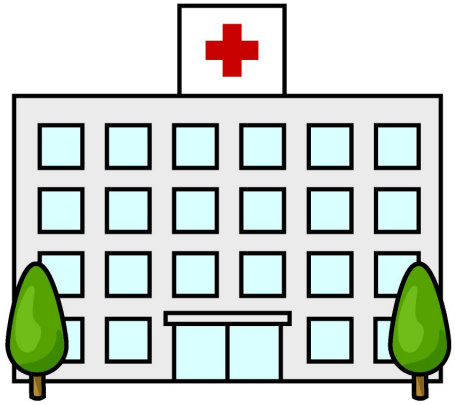
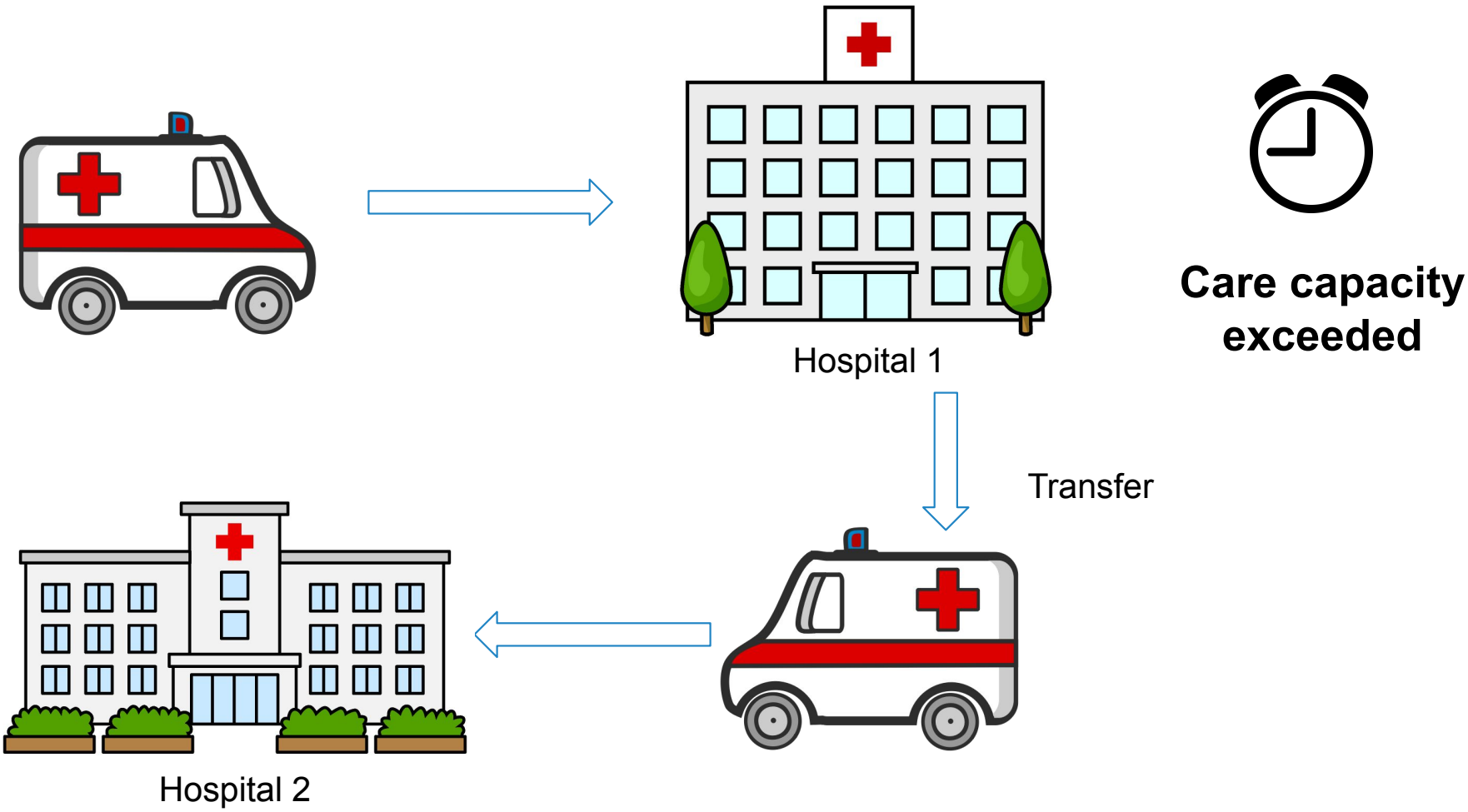


Hospital 1

Care capacity exceeded

EMS routing decisions impair COVID-19 care delivery



Current efforts to track hospital status do not support COVID-19 specific burden

County/Hospital Alert Tracking System (CHAT) in Maryland

Only accepting Priority I patient (emergently unstable)
No ECG monitored beds
Not accepting any patients

In urban environments with **multiple hospitals**, there is a need for EMS first responders to quickly select the **most appropriate hospital** for each suspected COVID-19 patient, based on the hospital's **COVID-19 care capacity**.

Design Requirements

Develop method for optimal hospital selection

Reduce time for patient to receive appropriate care

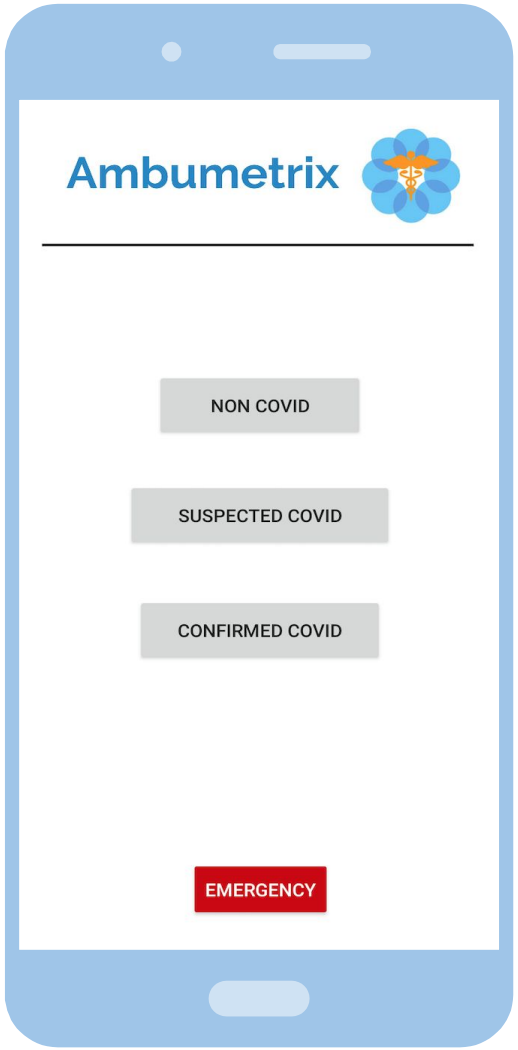
Preserve ED workflow

Our Solution: Ambumetrix

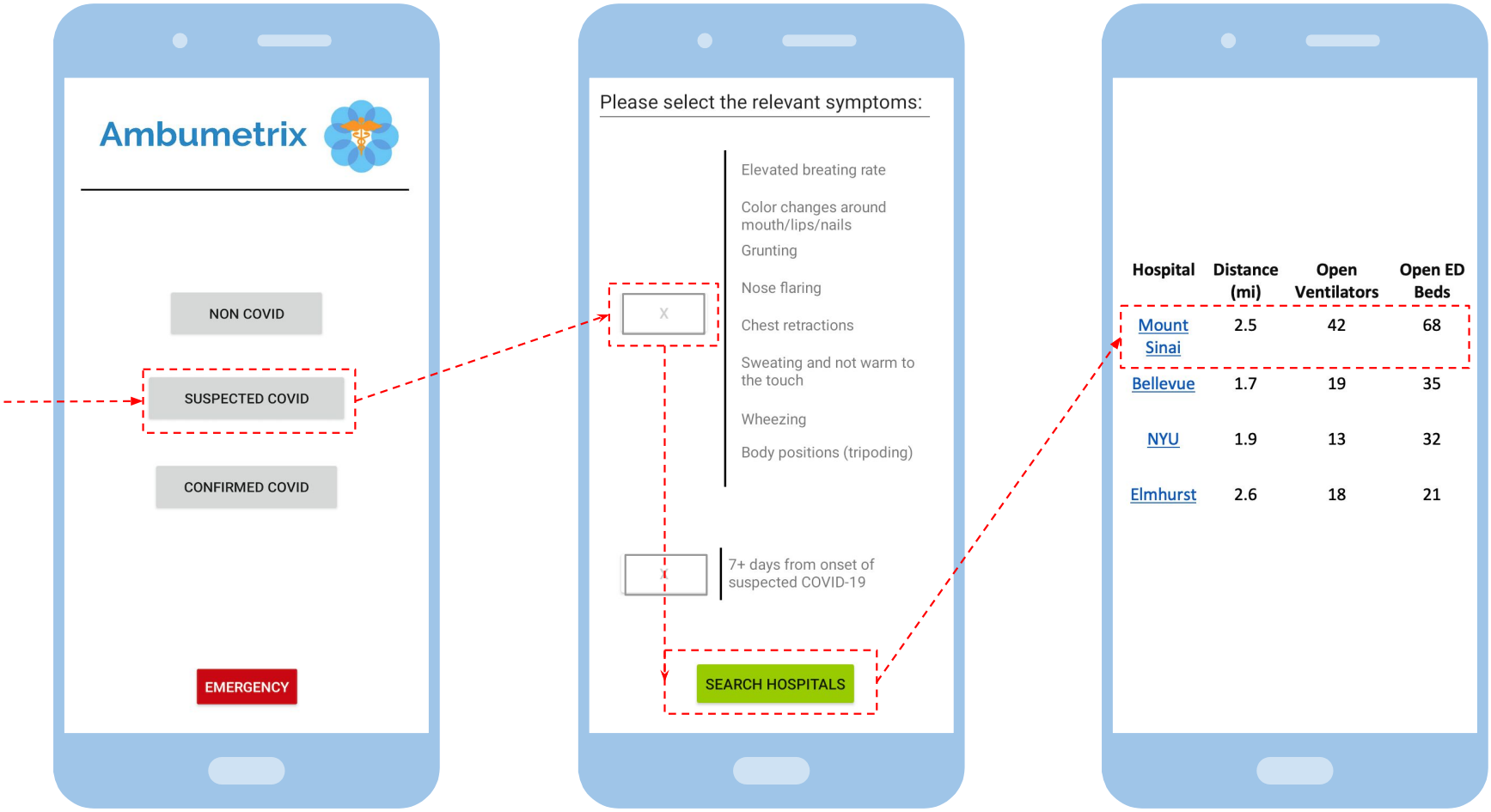
Mobile application for EMS users

Provides ranked hospital list according to highest capacity of care for a particular patient

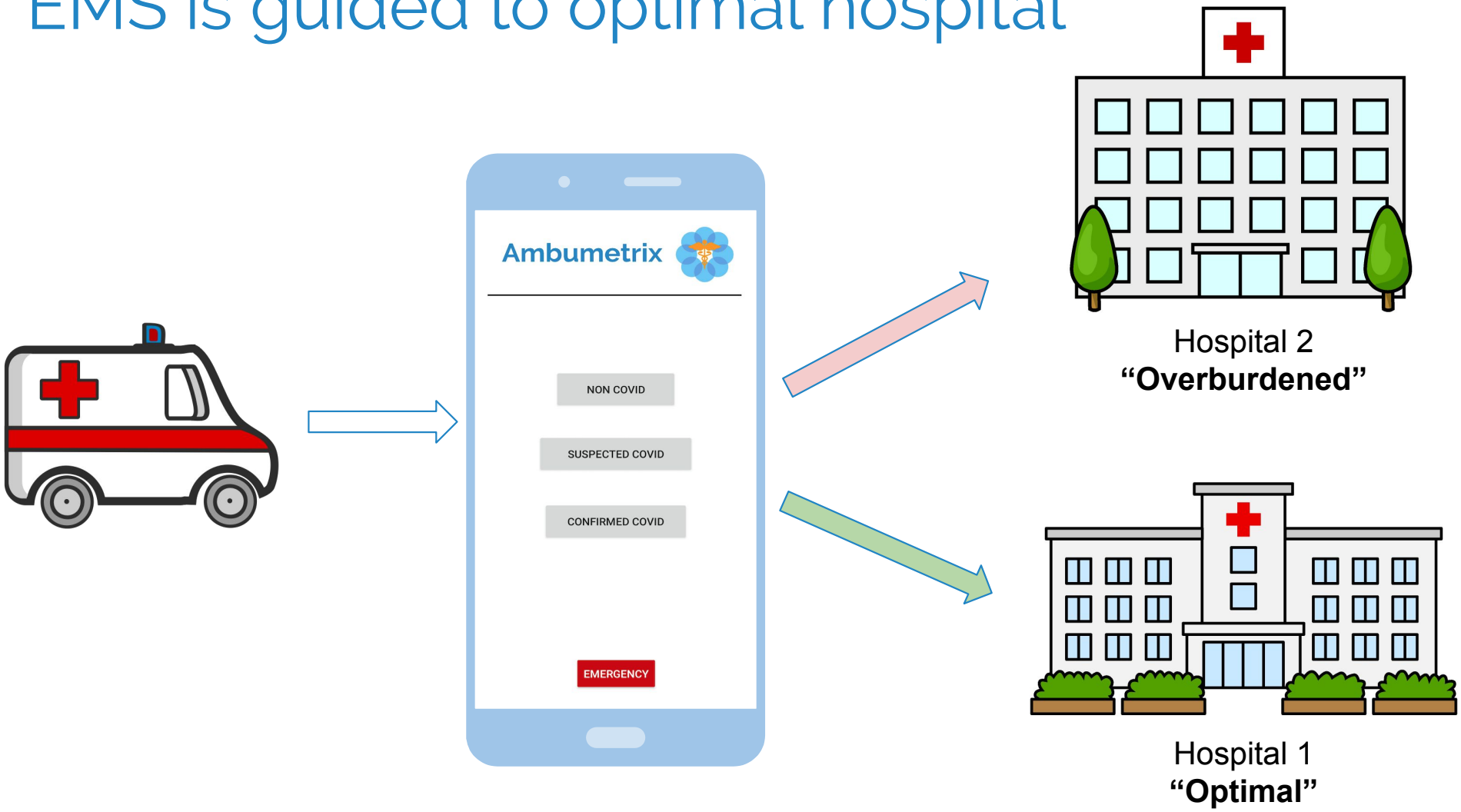
Allows for direct phone call to hospital of choice through app



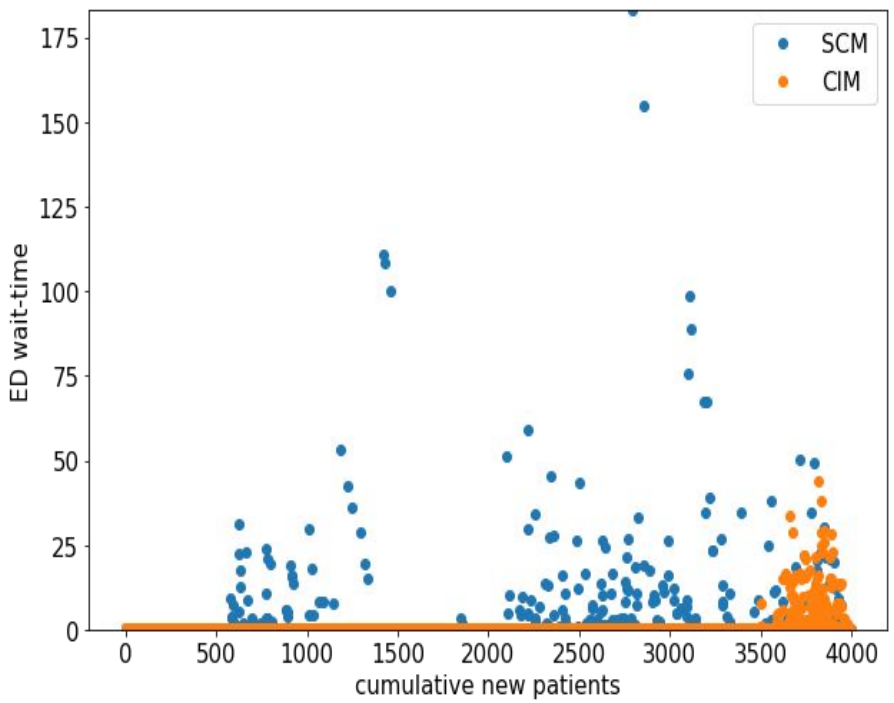
Ambumetrix Workflow

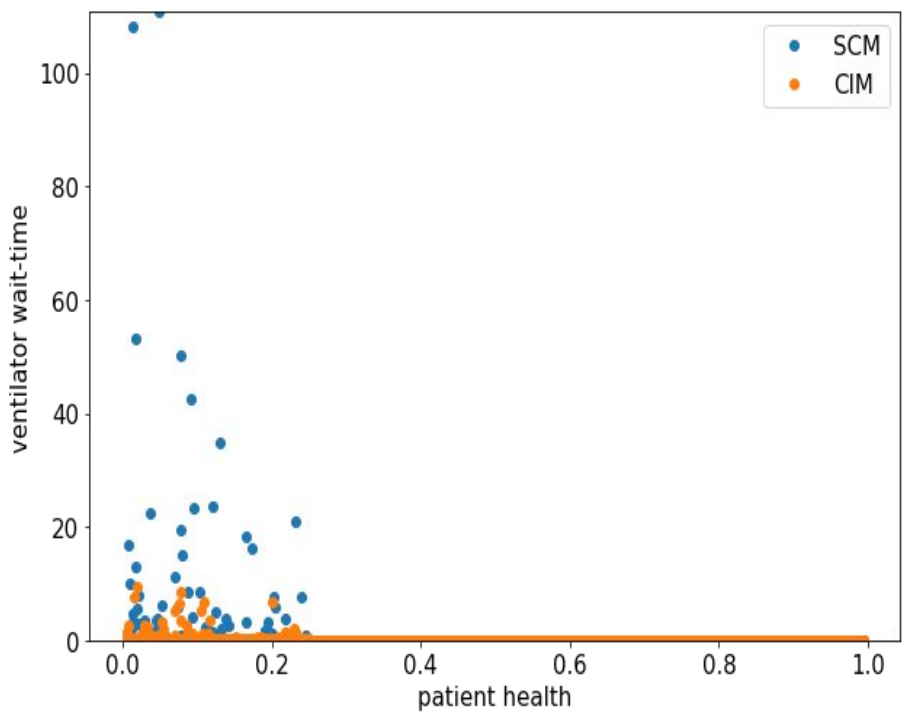


EMS is guided to optimal hospital



App decreases ED and ventilator wait time





Assumptions:

- ED beds: 20-60 per hospital
- ICU beds/ventilators: ⅓ of ED beds per hospital
- Probabilistic discharges

SCM = Standard of Care Model
CIM = Capacity Informed Model

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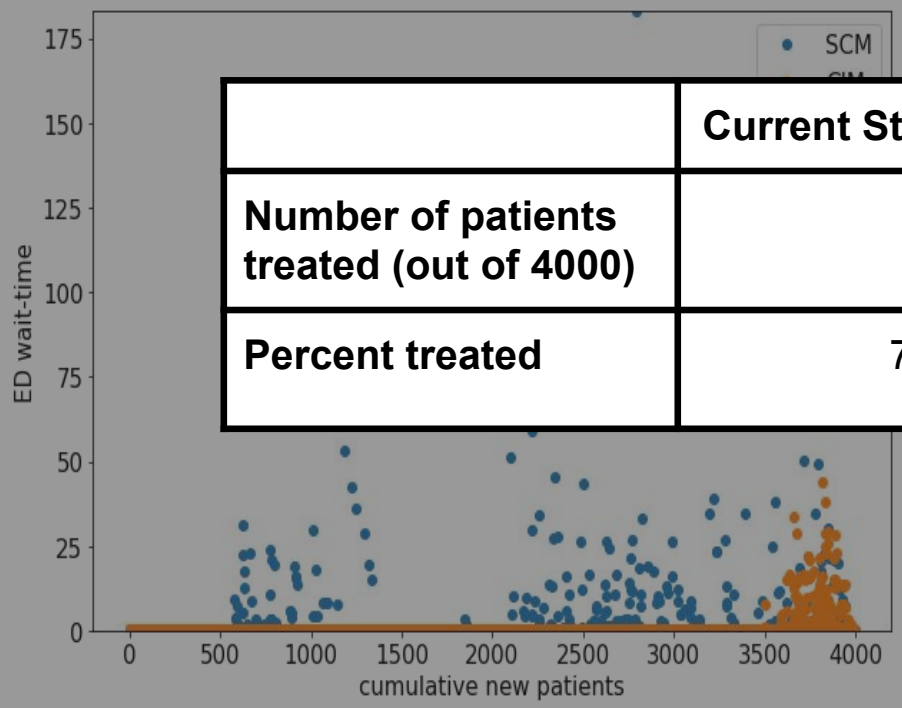

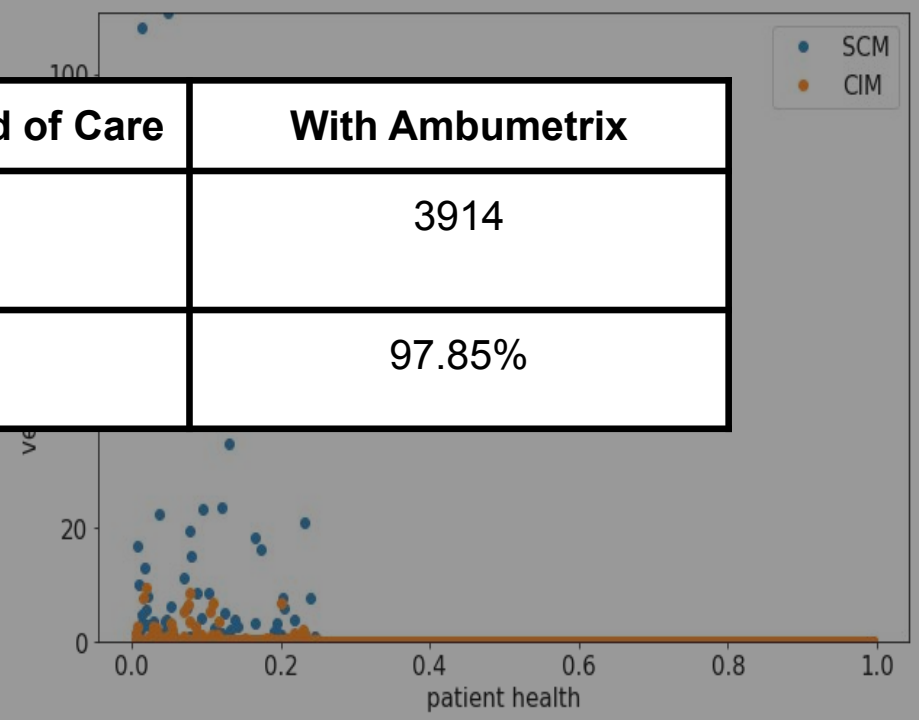

	Current Standard of Care	With Ambumetrix
Number of patients treated (out of 4000)	2964	3914
Percent treated	74.1%	97.85%

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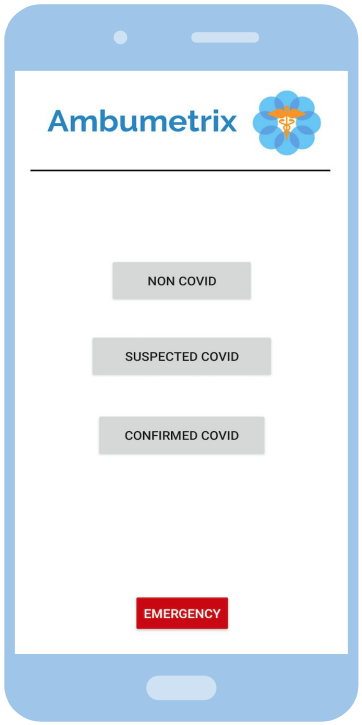
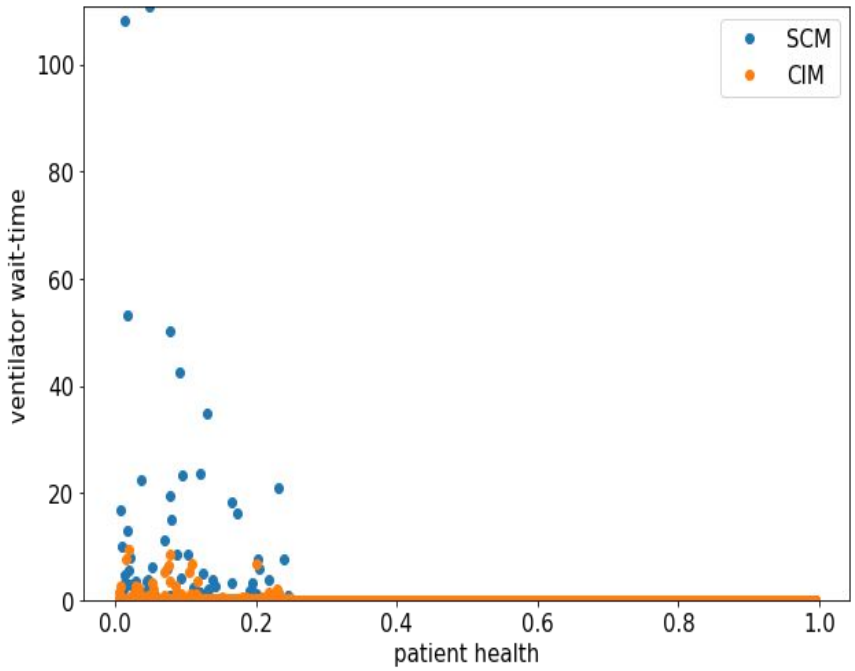
Large scale implementation in NY health systems

Develop beta	Partner with NY Dept. of Health	Pilot Study
Develop implementation with EPIC database	Coordinate connections across different hospital and EMS systems	Multi-center study for 1 week

All major NYC hospital systems are currently incentivized to improve ambulance coordination through the recently implemented **ET3 model.**

Summary

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